SurveyNCD: Design-Based Multimorbidity, Health-Inequality and Child-Growth Indicators from Complex Population Surveys

by Sujon Mia

Abstract Population health surveys such as the WHO STEPS surveys, the Demographic and Health Surveys and the Multiple Indicator Cluster Surveys record chronic conditions as self-reported answers and select respondents by stratified, clustered, unequal-probability designs. Turning those answers into defensible population estimates involves recoding inconsistent response codes, handling "don't know" answers, and carrying the design through to every standard error. The **SurveyNCD** package provides small, tested functions for this workflow: binary recoding, a self-reported multimorbidity count, design-based prevalence, a survey-weighted concentration index with design-based standard errors, categorisation of DHS-coded child growth z-scores, choropleth mapping and an exploratory weighted gradient-boosting module. We illustrate every function on simulated data drawn from a known population, validate the concentration index against **rineq** and a design bootstrap, and show in a 1,000-sample simulation that full-design intervals reach close to nominal coverage while analyses that ignore clustering do not.

1 Introduction

Multimorbidity, the presence of two or more chronic conditions in one person, is common and rising, and it is socially patterned (Barnett et al., 2012; Afshar et al., 2015). In low- and middle-income countries much of what is known about it comes from household surveys: the WHO STEPwise approach to non-communicable disease risk-factor surveillance (STEPS; Riley et al., 2016), the Demographic and Health Surveys (DHS; Corsi et al., 2012) and the Multiple Indicator Cluster Surveys (MICS; Khan and Hancioglu, 2019). These surveys share two features that shape the analysis. First, conditions are recorded as answers to questions such as "Has a doctor or health worker ever told you that you have high blood pressure?", coded in instrument-specific ways (1 = yes, 2 = no, 9 = don't know; or text). Second, respondents are selected by multi-stage designs with stratification, clustering and unequal selection probabilities, so that unweighted summaries are biased when the weights are informative and standard errors that ignore clustering are too small (Lumley, 2010).

The **survey** package (Lumley, 2004, 2010) provides the estimators these designs require. What it deliberately does not provide is the domain-specific layer that sits on top: deciding what a "don't know" does to a multimorbidity count, estimating subgroup prevalence without dropping rows from the design, ranking respondents by wealth for a concentration index when many share the same value, and decoding DHS conventions for anthropometric z-scores. In practice this layer is rewritten by hand for each project, and the mistakes it invites (missing answers counted as "no condition", subgroups analysed by subsetting the data frame, clustering ignored) leave no visible trace in the output.

SurveyNCD packages that layer as ten exported functions built on **survey**. This article describes the methods they implement and the design choices behind them, demonstrates every function on simulated data shipped with the package, and reports three kinds of validation: agreement with an independent implementation, robustness to the order of the data, and the frequency properties of the estimators over repeated samples from a known population.

2 Methods implemented

2.1 Design-based estimation

Let unit k in stratum h and primary sampling unit (PSU) c have sampling weight w_{hck}, the inverse of its inclusion probability. The prevalence of a binary indicator y is estimated by the ratio

$$\hat{p} = \frac{\sum_{h,c,k} w_{hck}\, y_{hck}}{\sum_{h,c,k} w_{hck}}, \tag{1}$$

and its variance by Taylor linearisation with the with-replacement approximation at the first stage (Binder, 1983; Lumley, 2010). Confidence intervals for proportions are computed on the logit scale by `survey::svyciprop(method = "logit")`, which keeps them inside $[0, 1]$. Subgroup (domain) estimates are obtained with `subset()` applied to the design object, which keeps all PSUs in the variance calculation; subsetting the data frame before building the design is a common error that understates the variance.

2.2 Self-reported multimorbidity

Given binary condition indicators $c_{i1}, \ldots, c_{iJ}$, the multimorbidity score is the weighted sum $\sum_j \omega_j c_{ij}$ with $\omega_j = 1$ by default, which is how the Functional Comorbidity Index (FCI; Groll et al., 2005) is scored. The FCI was developed for self-reported conditions; `fci_items()` returns its 18 conditions as a reference list. A respondent is classified as multimorbid when the unweighted count is at least two, the most common definition in prevalence studies (Fortin et al., 2012). When fewer than the 18 FCI conditions are available, the score is a condition count rather than the validated FCI, and the documentation says so.

2.3 Survey-weighted concentration index

The concentration index measures how a health variable y is distributed across the socioeconomic ranking (Wagstaff et al., 1991; O'Donnell et al., 2008):

$$C = \frac{2}{\mu} \operatorname{cov}_w(y_i, R_i), \tag{2}$$

where μ is the weighted mean of y and R_i the weighted fractional rank. For respondents sorted by wealth with cumulative weight $W_i = \sum_{j \le i} w_j$ and total W, the fractional rank is the mid-point of each respondent's interval of the cumulative distribution, $R_i = (W_{i-1} + w_i/2)/W$ (Lerman and Yitzhaki, 1989). When respondents are tied on the ranking variable, which is unavoidable with wealth quintiles, sorting places them in an arbitrary order and this formula then gives tied respondents different ranks. **SurveyNCD** instead assigns every member of a tied group g occupying the cumulative-weight interval $[L_g, U_g]$ the common rank $R_i = (L_g + U_g)/(2W)$, so the estimate depends only on the data and not on their order.

The point estimate and its standard error are obtained from the convenient regression of Kakwani et al. (1997),

$$2\sigma_R^2 \frac{y_i}{\mu} = \alpha + \beta R_i + \varepsilon_i, \tag{3}$$

where σ_R^2 is the weighted variance of R_i and $\hat{\beta} = C$. Fitting Equation 3 with `survey::svyglm()` gives a design-based sandwich standard error (Lumley and Scott, 2017). This standard error treats the ranks and μ as fixed; we assess the consequence of that simplification against a bootstrap that recomputes them (Section Validation).

2.4 Child anthropometry

DHS recode files store height-for-age, weight-for-age and weight-for-height z-scores multiplied by 100 (variables `hw70`, `hw71`, `hw72`) and use codes 9996–9999 for flagged or missing

measurements. `who_anthro_score()` rescales the values, sets the codes to missing, removes biologically implausible values using the WHO limits (height-for-age outside ± 6; weight-for-height outside ± 5; weight-for-age below -6 or above 5) and classifies the result as severe (< -3), moderate (-3 to < -2) or normal (WHO Multicentre Growth Reference Study Group, 2006). It does not compute z-scores from raw measurements; for that, the WHO **anthro** package is the reference implementation.

3 Related software

Table 1 positions **SurveyNCD** among related R packages. Comorbidity packages such as **comorbidity** (Gasparini, 2018) and **multimorbidity** (Bensken, 2023) map ICD-9 and ICD-10 diagnosis codes in claims or clinical records to Charlson, Elixhauser and related scores; they do not apply to self-reported survey answers and do not estimate population quantities. **rineq** (Devleesschauwer et al., 2025) computes relative, generalised and Erreygers-corrected (Erreygers, 2009) concentration indices with decomposition, using weights for the point estimate and an independence-based variance. **healthequal** (Antiporta et al., 2024) computes 21 summary measures of inequality from disaggregated subgroup estimates. **anthro** (Schumacher, 2026) computes WHO growth-standard z-scores and design-based prevalence from raw measurements. **SurveyNCD** overlaps with each of these at one point. Its contribution is to connect self-reported multimorbidity scoring, domain prevalence and a design-based concentration index in one workflow on **survey** design objects, not to provide new estimators.

Table 1: Scope of **SurveyNCD** and related R packages (as described in each package's documentation).

Package	Main input	Design-based SEs	Focus
survey	Respondent microdata	Yes	General estimators for complex designs
comorbidity	ICD-coded diagnoses	No	Charlson/Elixhauser scores
multimorbidity	ICD-coded claims	No	Comorbidity, multimorbidity and frailty measures
rineq	Respondent microdata	No (weights in point estimate)	Concentration indices and decomposition
healthequal	Subgroup estimates	From input estimates	21 summary measures of inequality
anthro	Raw child measurements	Yes	WHO z-scores and prevalence
SurveyNCD	Respondent microdata, raw codes	Yes	Self-reported multimorbidity, domain prevalence, design-based concentration index

4 Package design

Figure 1 shows how the ten functions fit together. Four decisions shape the package.

A thin layer over **survey**. All variance estimation is delegated to `svydesign()`, `svyciprop()`, `svymean()` and `svyglm()`. `mm_prevalence()` accepts column names and builds the design internally, while `survey_concentration_index()` accepts any `survey.design` object, including those made by **srvyr**. Both set `options(survey.lonely.psu = "adjust")` for the duration of the call, because single-PSU strata arise easily in subgroup analyses, and restore the user's setting on exit.

Missing answers are a stated policy. `recode_binary()` warns about any value it cannot place instead of guessing. `multimorbidity_index()` takes `na_action = "na"` (any missing condition gives a missing score) or `"ignore"` (sum the answered conditions); under either policy a respondent with every condition missing is scored as missing, never as healthy. `mm_prevalence()` reports how many respondents it excluded.

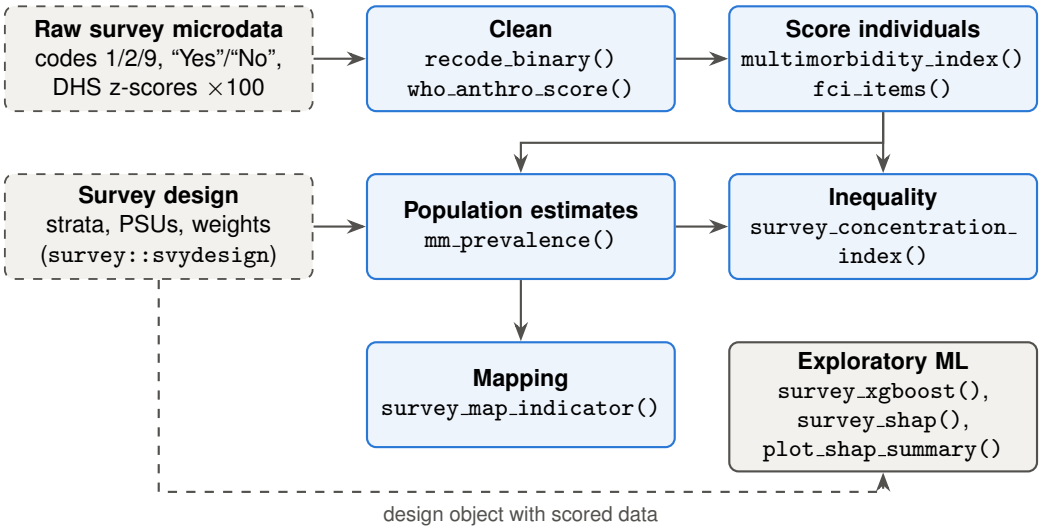


Figure 1: Workflow of **SurveyNCD**. Solid boxes are package functions; dashed boxes are inputs. Exploratory machine learning is shaded to mark that it does not provide design-based inference.

Results do not depend on the order of the data. This is the reason for the tie rule for fractional ranks described above.

Exploratory methods are fenced off. `survey_xgboost()` passes the sampling weights to **xgboost** (Chen and Guestrin, 2016) as case weights; `survey_shap()` extracts SHAP attributions (Lundberg and Lee, 2017) and `plot_shap_summary()` plots them. Strata and clusters play no part in any of these, so the documentation describes them as exploratory. **xgboost** and **sf** (Pebesma, 2018) are suggested rather than imported.

5 Example data

The package ships two simulated datasets so that every function can be run immediately. They are drawn from a synthetic population whose generating code is in 'data-raw/' of the source repository. Because the population is known, estimates can be compared with the truth.

The population has six regions of unequal size (60 to 300 enumeration areas each, 950 in total), each divided into urban and rural areas, giving 12 strata. It contains 142,133 households with one adult aged 18–69 each, and 64,169 children under five. Each adult's seven conditions are drawn from logistic models in age, sex, body mass index, household wealth and an area-level random effect. The wealth gradient is positive for hypertension and diabetes and negative for arthritis, asthma and depression. Child z-scores depend on wealth, age and an area effect. The sample selects ten areas by simple random sampling in every stratum and 20 households in each, so small regions and urban areas are over-sampled. Inclusion probabilities are exact, giving the design weights; these vary 15-fold (Table 2). Condition answers are coded as in STEPS with 1–4% "don't know" per item, and child z-scores follow DHS coding with flag codes and a few implausible values.

6 A worked analysis

All results in this section come from 'analysis/01_worked_example.R' in the repository. The code shown is abbreviated; the script runs in about ten seconds.

Table 2: The simulated example datasets shipped with SurveyNCD.

Item	Value
Adults (households)	2400
Children under 5	1034
Strata	12
Primary sampling units	120
Weight range (normalised)	0.21 to 3.08
Adults with any 'don't know' answer	346 (14.4%)

6.1 Cleaning and scoring

```
library("SurveyNCD")
library("survey")
options(survey.lonely.psu = "adjust")
d <- ncd_survey
d$htn <- recode_binary(d$htn_raw, yes = 1, no = 2)
d$dep <- recode_binary(d$dep_raw, yes = "Yes", no = "No")
table(raw = d$dep_raw, clean = d$dep, useNA = "ifany")
#>             clean
#> raw             0    1 <NA>
#>   Don't know    0    0   72
#>   No         2176    0    0
#>   Yes           0  152    0
```

After recoding the other five items the same way, `multimorbidity_index()` scores each respondent:

```
conds <- c("htn", "dm", "arth", "asthma", "angina", "stroke", "dep")
scored <- multimorbidity_index(d, conditions = conds, na_action = "na")
table(scored$mm_category, useNA = "ifany")
#>             None Single condition      Multimorbid             <NA>
#>             1310              557              187              346
```

The 346 respondents (14.4%) with at least one "don't know" receive a missing score instead of being counted as free of that condition.

6.2 Prevalence

Table 3 compares design-based estimates for each condition with unweighted sample percentages and with the true population values. Design effects of 1.1 to 2.7 show how much clustering inflates the variance relative to simple random sampling.

Table 3: Self-reported prevalence of each condition in the simulated survey: design-based estimates (strata, clusters, weights; logit 95% CI), unweighted sample percentages, the true value in the simulated population, and the design effect.

Condition	Don't know (n)	Design-based % (95% CI)	Unweighted %	Population truth %	Design effect
Hypertension	92	17.0 (14.8, 19.5)	16.5	15.8	2.36
Diabetes	68	7.1 (5.6, 9.1)	6.8	7.4	2.69
Arthritis	38	10.4 (9.0, 12.0)	9.9	10.1	1.47
Asthma	12	3.9 (3.1, 4.9)	4.0	4.1	1.27
Angina / heart disease	53	3.1 (2.2, 4.3)	3.0	3.2	1.98
Stroke	34	0.8 (0.5, 1.3)	1.1	1.2	1.09
Depression	72	6.5 (5.2, 8.1)	6.5	6.0	1.98

```
mm_prevalence(scored, ids = "psu", strata = "stratum", weights = "wt")
#> 346 of 2400 rows have a missing multimorbidity score and are excluded
```

```
#> from the survey-weighted estimate.
#>      n prevalence prevalence_se prevalence_lower prevalence_upper
#> 1 2054 0.09533923    0.01175125       0.07445529        0.1213131
#>   mean_conditions mean_conditions_se
#> 1       0.4850582         0.02954051
```

The design-based multimorbidity prevalence of 9.5% (95% CI 7.4–12.1%) matches the population value of 9.49%; the unweighted sample proportion is 9.1%. Subgroup estimates use the by argument (Table 4). Figure 2 shows them by wealth quintile and residence. With 2,000 respondents the intervals are wide and the gradient in multimorbidity is not monotone, which is consistent with the simulated population, where conditions with opposite wealth gradients largely cancel in the count.

Table 4: Design-based prevalence of multimorbidity (two or more of seven self-reported conditions) and mean number of conditions, from mm_prevalence(). 346 of 2400 respondents with any 'don't know' answer were excluded. True population values: multimorbidity 9.5%, mean 0.48 conditions.

Characteristic	Group	n	Multimorbidity % (95% CI)	Mean conditions (SE)
All adults		2054	9.5 (7.4, 12.1)	0.49 (0.030)
Sex	Female	1050	9.9 (7.4, 13.2)	0.50 (0.037)
	Male	1004	9.1 (6.8, 12.2)	0.47 (0.036)
Residence	Rural	1038	9.3 (6.6, 12.9)	0.48 (0.038)
	Urban	1016	10.1 (7.5, 13.5)	0.49 (0.044)
Wealth quintile	Q1 (poorest)	399	8.5 (5.0, 13.9)	0.46 (0.054)
	Q2	418	9.4 (6.1, 14.1)	0.48 (0.044)
	Q3	404	11.9 (7.6, 18.3)	0.52 (0.061)
	Q4	396	5.9 (3.7, 9.4)	0.42 (0.039)
	Q5 (richest)	437	11.9 (8.2, 17.1)	0.55 (0.051)
Region	Region A	341	10.1 (6.6, 15.2)	0.48 (0.054)
	Region B	343	10.2 (4.7, 20.7)	0.49 (0.091)
	Region C	342	10.6 (6.6, 16.6)	0.56 (0.042)
	Region D	343	4.2 (2.3, 7.7)	0.35 (0.037)
	Region E	336	10.5 (6.4, 16.7)	0.51 (0.074)
	Region F	349	10.9 (7.0, 16.4)	0.55 (0.059)



Figure 2: Design-based multimorbidity prevalence by wealth quintile, overall and by residence, with logit 95% confidence intervals (simulated data).

6.3 Inequality

```
des <- svydesign(ids = ~psu, strata = ~stratum, weights = ~wt,
                 data = scored, nest = TRUE)
survey_concentration_index(des, outcome = dm, wealth = wealth_score)
#> # A tibble: 1 x 7
#>   Concentration_Index Standard_Error Lower_CI Upper_CI  p_value Outcome_Mean
#>                 <dbl>          <dbl>    <dbl>    <dbl>    <dbl>        <dbl>
#> 1               0.314         0.0547    0.206    0.421  1.00e-8       0.0714
#> # i 1 more variable: n <int>
```

Table 5 and Figure 3 report concentration indices for each condition, the condition count, multimorbidity and child undernutrition. Diabetes and hypertension are concentrated among richer adults and depression among poorer adults, as built into the simulation. Because these gradients cancel, the concentration index for the condition count is close to zero (true value 0.028), which illustrates that a count can hide opposing inequalities in its components.

Table 5: Wealth-related concentration indices from survey_concentration_index() (ranking: continuous wealth score). Negative values: concentrated among poorer people; positive: among richer people. Design-based SEs account for strata and clusters.

Outcome	Population	CI (95% CI)	SE	p	n
Hypertension	Adults	0.123 (0.065, 0.181)	0.030	<0.001	2308
Diabetes	Adults	0.314 (0.206, 0.421)	0.055	<0.001	2332
Arthritis	Adults	-0.083 (-0.168, 0.001)	0.043	0.052	2362
Asthma	Adults	-0.057 (-0.189, 0.076)	0.068	0.401	2388
Angina / heart disease	Adults	0.019 (-0.129, 0.166)	0.075	0.805	2347
Stroke	Adults	-0.199 (-0.503, 0.104)	0.155	0.198	2366
Depression	Adults	-0.153 (-0.266, -0.040)	0.058	0.008	2328
Number of conditions	Adults	0.025 (-0.028, 0.079)	0.027	0.352	2054
Multimorbidity	Adults	0.036 (-0.070, 0.142)	0.054	0.502	2054
Stunting	Children	-0.250 (-0.312, -0.188)	0.032	<0.001	1015
Underweight	Children	-0.342 (-0.432, -0.252)	0.046	<0.001	1009
Wasting	Children	-0.210 (-0.374, -0.046)	0.084	0.012	1007

6.4 Child undernutrition

```
ch <- child_anthro
ch$stunting <- who_anthro_score(ch$hw70, indicator = "stunting")
table(ch$stunting, useNA = "ifany")
#>   Severe stunting Moderate stunting   Normal stunting              <NA>
#>                81               209               725                19
```

Table 6 summarises the three indicators after flag codes and implausible values are removed. In this particular sample the stunting interval (26.9–34.1%) narrowly misses the population value of 26.8%. That is the kind of event a 95% interval should produce about once in twenty samples, and the simulation study below shows its long-run coverage. Stunting is strongly concentrated among poorer children ($C = -0.250$, 95% CI -0.312 to -0.188; population value -0.259).

Table 6: Child undernutrition from DHS-coded z-scores using who_anthro_score(): flag codes and biologically implausible values set to missing, then design-based prevalence.

Indicator	DHS flag codes (n)	Implausible removed (n)	Analysed (n)	Severe %	Moderate %	Moderate or severe % (95% CI)
Stunting (HAZ < -2)	12	7	1015	8.9	21.5	30.4 (26.9, 34.1)
Underweight (WAZ < -2)	19	6	1009	4.0	14.2	18.2 (14.7, 22.3)
Wasting (WHZ < -2)	21	6	1007	0.7	6.8	7.5 (5.8, 9.7)

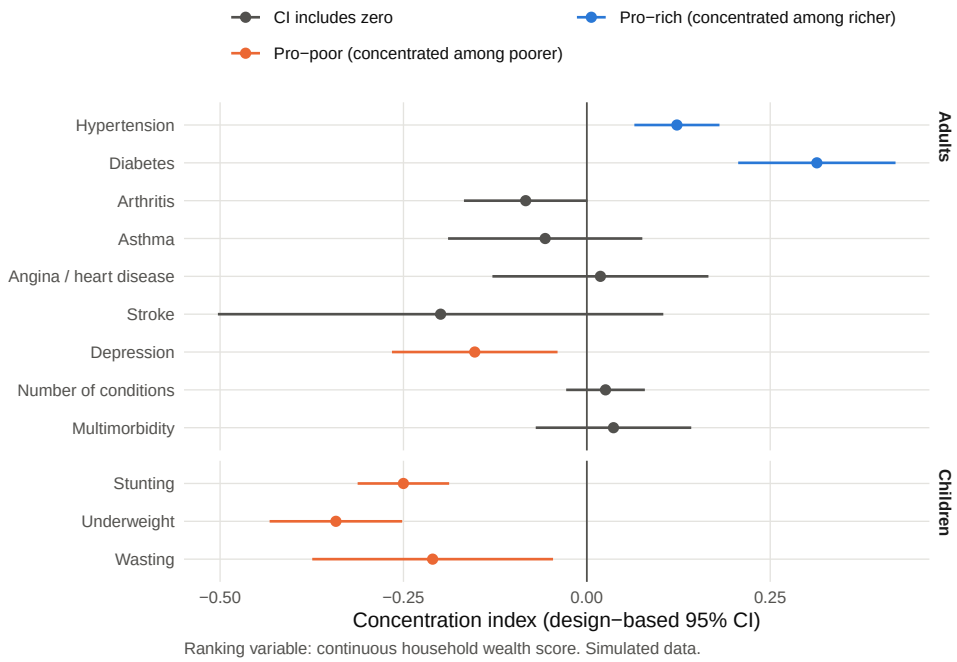


Figure 3: Wealth-related concentration indices with design-based 95% confidence intervals (simulated data). Colour marks the direction when the interval excludes zero.

6.5 Mapping and exploratory modelling

Regional estimates from `mm_prevalence(by = "region")` are joined to an **sf** object and drawn by `survey_map_indicator()` (Figure 4). Because the regions are fictitious, the example uses a schematic grid of six polygons; with real data, any boundary file with a matching key column works.

Finally, a weighted gradient-boosting model for the number of conditions with age, BMI, wealth, sex and residence as features ranks age as by far the most influential feature (mean absolute SHAP value 0.21), followed by BMI (0.08) and wealth (0.07); Figure 5. These attributions describe the fitted model, not causal effects, and carry no design-based uncertainty.

7 Validation

7.1 Agreement with an independent implementation

Table 7 compares `survey_concentration_index()` with `rineq::ci()` on the same weighted data. The wealth score is recorded to three decimals and has tied values. **rineq** ranks ties by their position in the data while **SurveyNCD** gives them a common rank, so the two differ by up to 4×10^{-5}. With the ties broken by a negligible perturbation, they agree to within 5×10^{-16}. For standard errors, the design-based value from **SurveyNCD** is within 11% of a 500-replicate bootstrap that resamples PSUs within strata and recomputes the ranks and the mean in every replicate. This supports treating the ranks as fixed in Equation 3 at this sample size. The independence-based standard error from **rineq** is 9–27% smaller, as expected when clustering is ignored.

7.2 Order invariance

Version 0.1.0 of **SurveyNCD** assigned tied respondents ranks by row order. Table 8 reports the concentration index of the condition count ranked by wealth quintile over 50 random

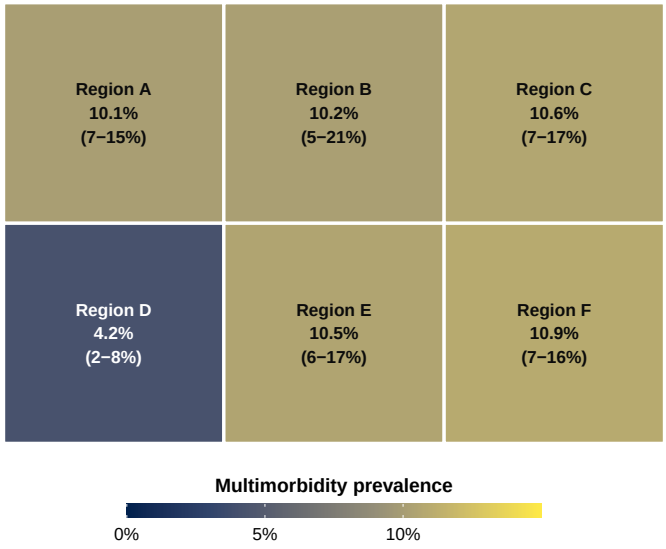


Figure 4: Multimorbidity prevalence by region, drawn with `survey_map_indicator()` on a schematic map of six fictitious regions. Labels give estimates and 95% confidence intervals.

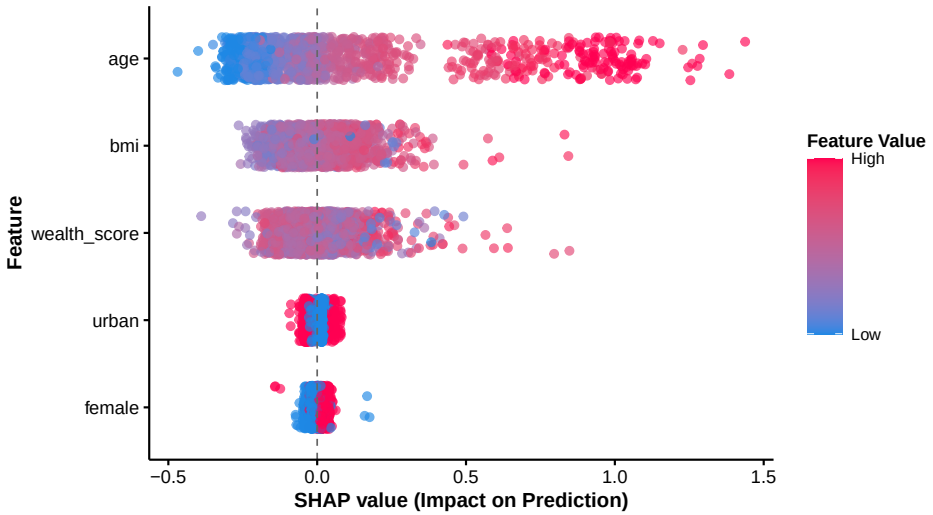


Figure 5: SHAP summary for a survey-weighted gradient-boosting model of the number of conditions (`survey_xgboost()`, `survey_shap()`, `plot_shap_summary()`; 300 rounds, depth 3). Exploratory: case weights only.

Table 7: Concentration index (adults with complete answers on all seven conditions, n = 2054) from survey_concentration_index() compared with rineq::ci() (version 0.0.2) and with a 500-replicate design bootstrap that re-computes the fractional ranks in every replicate. The wealth score has 379 tied values; rineq ranks ties by row order while SurveyNCD gives tied respondents a common mid-rank, which explains the small differences; with ties broken the two agree to rounding error. rineq's standard error assumes independent observations and ignores strata and clusters.

Outcome	SurveyNCD CI	rineq CI	Abs. diff. (ties)	Abs. diff. (no ties)	SE: SurveyNCD (design)	SE: bootstrap (design)	SE: rineq (iid)
Hypertension	0.103	0.103	4.2e-05	8.3e-17	0.032	0.033	0.029
Diabetes	0.320	0.320	1.9e-05	3.3e-16	0.060	0.056	0.044
Depression	-0.194	-0.194	2.3e-05	1.1e-16	0.063	0.057	0.048
Number of conditions	0.025	0.025	2.4e-05	4.9e-17	0.027	0.026	0.020
Multimorbidity	0.036	0.036	3.9e-05	4.7e-16	0.054	0.052	0.040

orderings of the same data. Under the old rule the estimate ranged from 0.011 to 0.029 around a reference value of 0.020, a spread almost as large as the estimate. The current rule returns 0.020 every time. A regression test guards this behaviour.

Table 8: Concentration index of the number of conditions ranked by wealth quintile (five tied groups), over 50 random row orders of the same data.

Version	Min CI	Max CI	Range	Reference (mid-rank)
0.1.0 (row-order ranks)	0.011	0.029	0.018	0.020
0.2.0 (mid-ranks for ties)	0.020	0.020	4.9e-16	0.020

7.3 Simulation study

To assess bias and interval coverage, we drew 1,000 independent samples from the population with the same design and analysed each in three ways: the full design (strata, PSUs, weights) as **SurveyNCD** does; weights only, ignoring strata and clusters; and an unweighted analysis assuming independent observations. Following Morris et al. (2019), Table 9 reports relative bias, the ratio of the mean estimated standard error to the empirical standard deviation of the estimates, and coverage of nominal 95% intervals. The Monte Carlo standard error of coverage is about 0.7 percentage points.

Table 9: Monte Carlo evaluation over 1000 independent samples drawn with the same stratified two-stage design from one simulated population (142,133 adults; 64,169 children). Coverage of nominal 95% intervals (Monte Carlo SE about 0.7 points). SE ratio = mean estimated SE / empirical SE.

Estimand	Method	True value	Mean estimate	Rel. bias %	Empirical SE	Mean model SE	SE ratio	Coverage %
Multimorbidity prevalence	Full design (SurveyNCD)	0.0949	0.0953	0.4	0.0093	0.0094	1.01	95.5
	Weights only		0.0953	0.4	0.0093	0.0078	0.84	90.7
	Unweighted, iid		0.0959	1.1	0.0075	0.0065	0.87	90.5
CI: number of conditions	Full design (SurveyNCD)	0.0275	0.0284	2.9	0.0261	0.0260	0.99	94.2
	Weights only		0.0284	2.9	0.0261	0.0236	0.90	92.6
	Unweighted, iid		0.0331	20.0	0.0223	0.0200	0.89	91.6
Stunting prevalence	Full design (SurveyNCD)	0.2684	0.2676	-0.3	0.0206	0.0199	0.97	93.4
	Weights only		0.2676	-0.3	0.0206	0.0165	0.80	88.7
	Unweighted, iid		0.2642	-1.6	0.0159	0.0135	0.85	88.9
CI: stunting	Full design (SurveyNCD)	-0.2589	-0.2591	-0.1	0.0349	0.0350	1.00	95.0
	Weights only		-0.2591	-0.1	0.0349	0.0333	0.95	93.9
	Unweighted, iid		-0.2657	-2.6	0.0288	0.0278	0.97	94.0

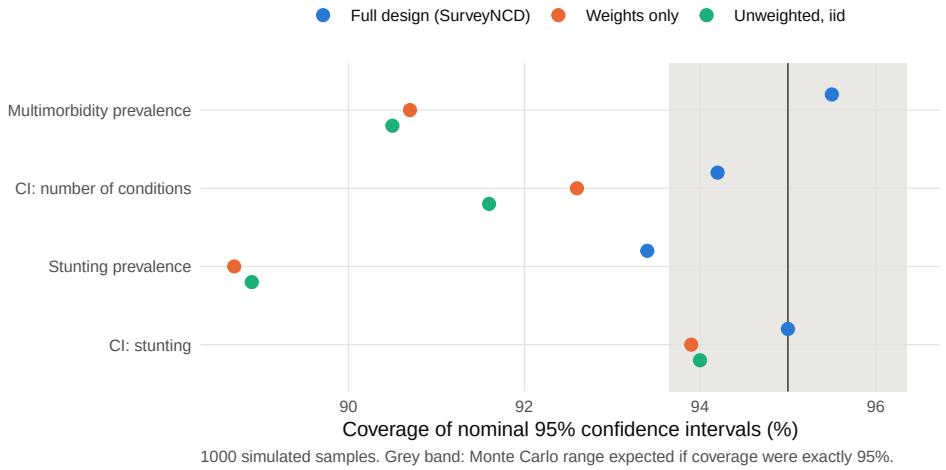


Figure 6: Coverage of nominal 95% confidence intervals over 1,000 simulated samples. The grey band is the range expected from Monte Carlo error alone if coverage were exactly 95%.

With the full design, bias was below 3% for all four estimands, standard-error ratios were 0.97–1.01 and coverage was 93.4–95.5% (Figure 6). Ignoring strata and clusters left the point estimates unchanged but understated the standard errors by up to 20%, and coverage

fell to 88.7–93.9%. The unweighted analysis also understated uncertainty and inflated the concentration index of the condition count by 20%, because urban and small-region households were over-sampled. Full-design coverage for stunting prevalence (93.4%) was slightly below nominal; with only ten PSUs per stratum, a t reference distribution or a design-degrees-of-freedom correction would be a natural refinement.

The whole study takes about five minutes on one core ('analysis/03_simulation_study.R').

8 Limitations

Complete-case exclusion under `na_action = "na"` is unbiased only when "don't know" answers are missing completely at random; in the example it removed 14% of respondents, and no weight adjustment is made. `mm_prevalence()` and `survey_concentration_index()` take one grouping variable at a time. The concentration-index standard error treats ranks as fixed and uses a normal reference distribution. The validation above suggests this works at typical survey sizes, but it may not with few PSUs. `who_anthro_score()` covers undernutrition tiers only. The gradient-boosting module offers no design-based inference. The simulated population is a deliberately simple model and is not a substitute for validation against published estimates from real surveys, which we plan as the package is applied to STEPS and DHS microdata.

9 Summary

SurveyNCD turns raw self-reported survey answers into design-based estimates of multimorbidity, its socioeconomic distribution and child undernutrition, using **survey** for all variance estimation. Shipping simulated data with a known population, and the scripts that reproduce every table and figure here, lets users check what the functions do before applying them to their own surveys. The package is on CRAN, and development takes place at https://github.com/StatAid-Research-Lab/SurveyNCD.

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Use of generative AI. [TODO, AUTHOR TO COMPLETE AND CONFIRM: Claude Sonnet 5.5 (Anthropic) was used to help write the package's R code. Claude (Anthropic) was used in October 2026 to help draft and edit this article and to write the simulation, validation and figure scripts. State the nature of the assistance with the code. Add the sentence that the author reviewed, ran and verified all code and results and takes responsibility for them, but only if this is true.]

References

S. Afshar, P. J. Roderick, P. Kowal, B. D. Dimitrov, and A. G. Hill. Multimorbidity and the inequalities of global ageing: A cross-sectional study of 28 countries using the World Health Surveys. *BMC Public Health*, 15:776, 2015. doi: 10.1186/s12889-015-2008-7. [p1]

D. A. Antiporta, P. Menéndez, K. Kirkby, and A. Hosseinpoor. *healthequal: Compute Summary Measures of Health Inequality*, 2024. URL https://CRAN.R-project.org/package=healthequal. R package version 1.0.1. [p3]

K. Barnett, S. W. Mercer, M. Norbury, G. Watt, S. Wyke, and B. Guthrie. Epidemiology of multimorbidity and implications for health care, research, and medical education: A cross-sectional study. *The Lancet*, 2012. doi: 10.1016/S0140-6736(12)60240-2. [p1]

W. Bensken. *multimorbidity: Harmonizing Various Comorbidity, Multimorbidity, and Frailty Measures*, 2023. URL https://CRAN.R-project.org/package=multimorbidity. R package version 0.5.1. [p3]

D. A. Binder. On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review*, 51(3):279–292, 1983. doi: 10.2307/1402588. [p2]

T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785. [p4]

D. J. Corsi, M. Neuman, J. E. Finlay, and S. V. Subramanian. Demographic and health surveys: A profile. *International Journal of Epidemiology*, 41(6):1602–1613, 2012. doi: 10.1093/ije/dys184. [p1]

B. Devleesschauwer, S. Willimès, C. Van Malderen, P. Konings, N. Speybroeck, and K. Meili. *rineq: Concentration Index and Decomposition for Health Inequalities*, 2025. URL https://CRAN.R-project.org/package=rineq. R package version 0.3.0. [p3]

G. Erreygers. Correcting the concentration index. *Journal of Health Economics*, 28(2):504–515, 2009. doi: 10.1016/j.jhealeco.2008.02.003. [p3]

M. Fortin, M. Stewart, M.-E. Poitras, J. Almirall, and H. Maddocks. A systematic review of prevalence studies on multimorbidity: Toward a more uniform methodology. *Annals of Family Medicine*, 10(2):142–151, 2012. doi: 10.1370/afm.1337. [p2]

A. Gasparini. comorbidity: An R package for computing comorbidity scores. *Journal of Open Source Software*, 3(23):648, 2018. doi: 10.21105/joss.00648. [p3]

D. L. Groll, T. To, C. Bombardier, and J. G. Wright. The development of a comorbidity index with physical function as the outcome. *Journal of Clinical Epidemiology*, 58(6):595–602, 2005. doi: 10.1016/j.jclinepi.2004.10.018. [p2]

N. Kakwani, A. Wagstaff, and E. van Doorslaer. Socioeconomic inequalities in health: Measurement, computation, and statistical inference. *Journal of Econometrics*, 77(1):87–103, 1997. doi: 10.1016/S0304-4076(96)01807-6. [p2]

S. Khan and A. Hancioglu. Multiple indicator cluster surveys: Delivering robust data on children and women across the globe. *Studies in Family Planning*, 50(3):279–286, 2019. doi: 10.1111/sifp.12103. [p1]

R. I. Lerman and S. Yitzhaki. Improving the accuracy of estimates of Gini coefficients. *Journal of Econometrics*, 1989. doi: 10.1016/0304-4076(89)90074-2. [p2]

T. Lumley. Analysis of complex survey samples. *Journal of Statistical Software*, 9(8):1–19, 2004. doi: 10.18637/jss.v009.i08. [p1]

T. Lumley. *Complex Surveys: A Guide to Analysis Using R*. John Wiley & Sons, Hoboken, NJ, 2010. doi: 10.1002/9780470580066. [p1, 2]

T. Lumley and A. Scott. Fitting regression models to survey data. *Statistical Science*, 2017. doi: 10.1214/16-STS605. [p2]

S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, 2017. URL https://arxiv.org/abs/1705.07874. [p4]

T. P. Morris, I. R. White, and M. J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi: 10.1002/sim.8086. [p10]

O. O'Donnell, E. van Doorslaer, A. Wagstaff, and M. Lindelow. *Analyzing Health Equity Using Household Survey Data: A Guide to Techniques and Their Implementation*. The World Bank, Washington, DC, 2008. doi: 10.1596/978-0-8213-6933-3. [p2]

E. Pebesma. Simple features for R: Standardized support for spatial vector data. *The R Journal*, 10(1):439–446, 2018. doi: 10.32614/RJ-2018-009. [p4]

L. Riley, R. Guthold, M. Cowan, S. Savin, L. Bhatti, T. Armstrong, and R. Bonita. The World Health Organization STEPwise approach to noncommunicable disease risk-factor surveillance: Methods, challenges, and opportunities. *American Journal of Public Health*, 106(1):74–78, 2016. doi: 10.2105/AJPH.2015.302962. [p1]

D. Schumacher. *anthro: Computation of the WHO Child Growth Standards*, 2026. URL https://CRAN.R-project.org/package=anthro. R package version 1.1.0. [p3]

A. Wagstaff, P. Paci, and E. van Doorslaer. On the measurement of inequalities in health. *Social Science & Medicine*, 33(5):545–557, 1991. doi: 10.1016/0277-9536(91)90212-U. [p2]

WHO Multicentre Growth Reference Study Group. WHO child growth standards based on length/height, weight and age. *Acta Paediatrica*, 95(S450):76–85, 2006. doi: 10.1111/j.1651-2227.2006.tb02378.x. [p3]

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